SURV/SURVMETH 720

Total Survey Error

TSE and Found Data

Class #12 – Final class!

December 1, 2025

Tradeoff/Interplay between Error Sources

- Design features may affect more than one source of error.
- TSE perspective: base design decisions on overall effects - not just their effects on one source of error.
- Since bias is directional, individual errors can:
  - accentuate each other, such that net bias > sum of absolute biases
  - be independent
  - cancel each other out, such that net bias ≤ sum of absolute biases
- Illustration: https://bit.ly/3gg0N10

TSE and found data

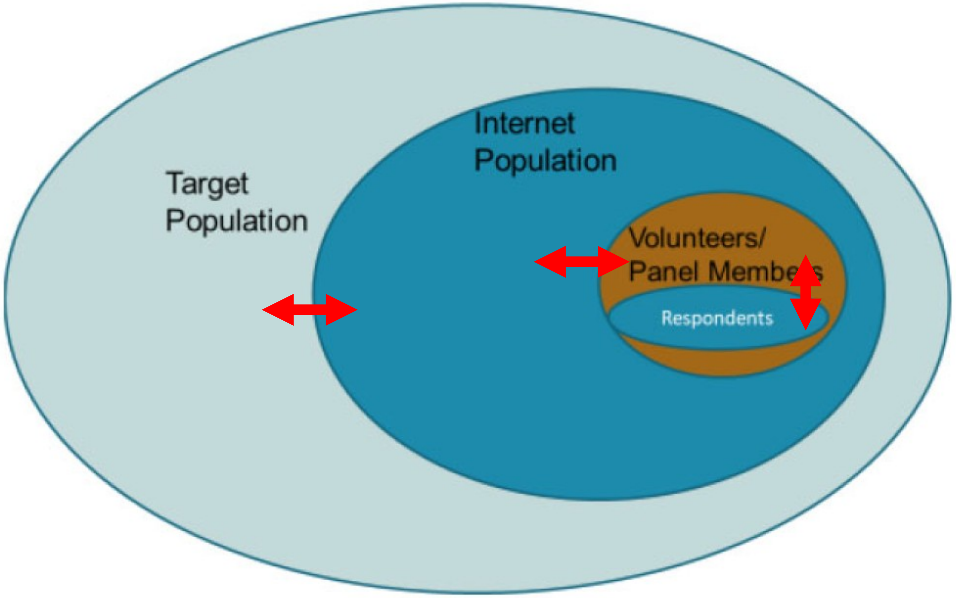
Designed vs. Found Data
(Groves, 2011)

- **Designed** data: designed and produced by researchers
  - Surveys, experiments, qualitative interviews
- **Found** data: data that already exist as a by-product of something else; exist independently of a research design (non-reactive)
- Salganik (2018) makes a similar distinction: **custommade** versus **readymade**

Non-Probability Panels

- Increased use of nonprobability Internet panels
  - No sampling frame per se
- How should we think about representation error in this setting?

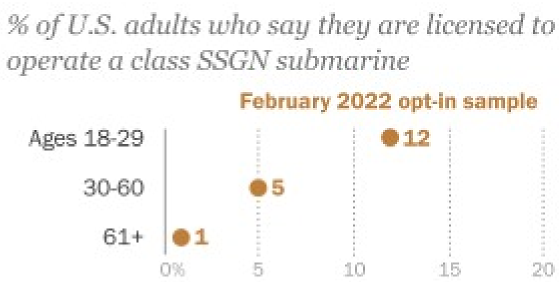
Tourangeau (2020): Three Forms of Error

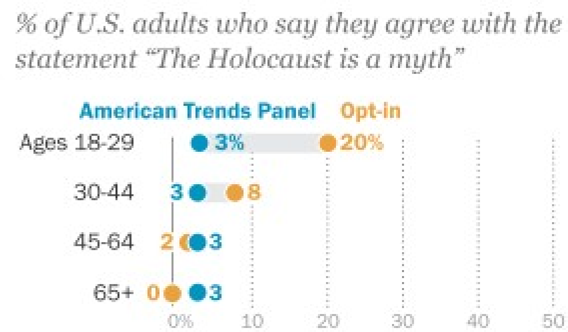


- Coverage bias: difference between the Internet population and the target population
- Selection bias: difference between the panelists selected for a given survey and the Internet population
- Nonresponse bias: difference between the panelists asked to complete a specific survey and the ones who actually do

Side note…

(Mercer et al. 2024 and Kennedy et al. 2024)





– About a quarter (24%) of opt-in cases claiming to be Hispanic said they were licensed to operate a nuclear sub, versus 2% of non-Hispanics.

This study also raises the question of whether Hispanic adults are more prone to providing bogus answers than other adults. We do not think that is a credible explanation for the results observed. To our minds, realizing that many of these "Hispanic" cases are, in all likelihood, not actually Hispanic is the single most important finding. Its implications for survey statisticians are profound. The implication is that we should not be relying exclusively on techniques like sample matching, propensity models, or hierarchical regressions to fix errors in commercial nonprobability samples. All such approaches assume that respondents are who they say they are; that their demographic information is measured with little to no error. This study demonstrates that with certain types of nonprobability data (namely commercial, online panels), that is not a safe assumption.

Found Data: Potential Strengths

- **Huge volume**
  - Helpful for studying: rare events, small groups, and small differences
- **High velocity**
  - Helpful for studying: events in real-time, unexpected events, trends
- **Nonreactivity**
  - Participants may not be aware their data are being captured
  - Helpful for studying: sensitive behavior and attitudes

Salganik (2018)
https://www.bitbybitbook.com

Potential Strengths…

- **Unobtrusive**
  – Doesn’t require active (potentially burdensome) data collection
- **Relatively cheap**
  – Data already exist

Found Data: Potential Limitations

- **‘Nonrepresentative’**
  - problematic for generalizing to larger target populations
  - demographics, behavior on other platforms, data to operationalize theoretical constructs

- **Online system engineering and algorithms**
  - “algorithmic confounding”

Found Data: Potential Limitations…

- **Volatility**
  - constantly-changing online systems, user groups, user behaviors
  - problematic for studying behavior over time

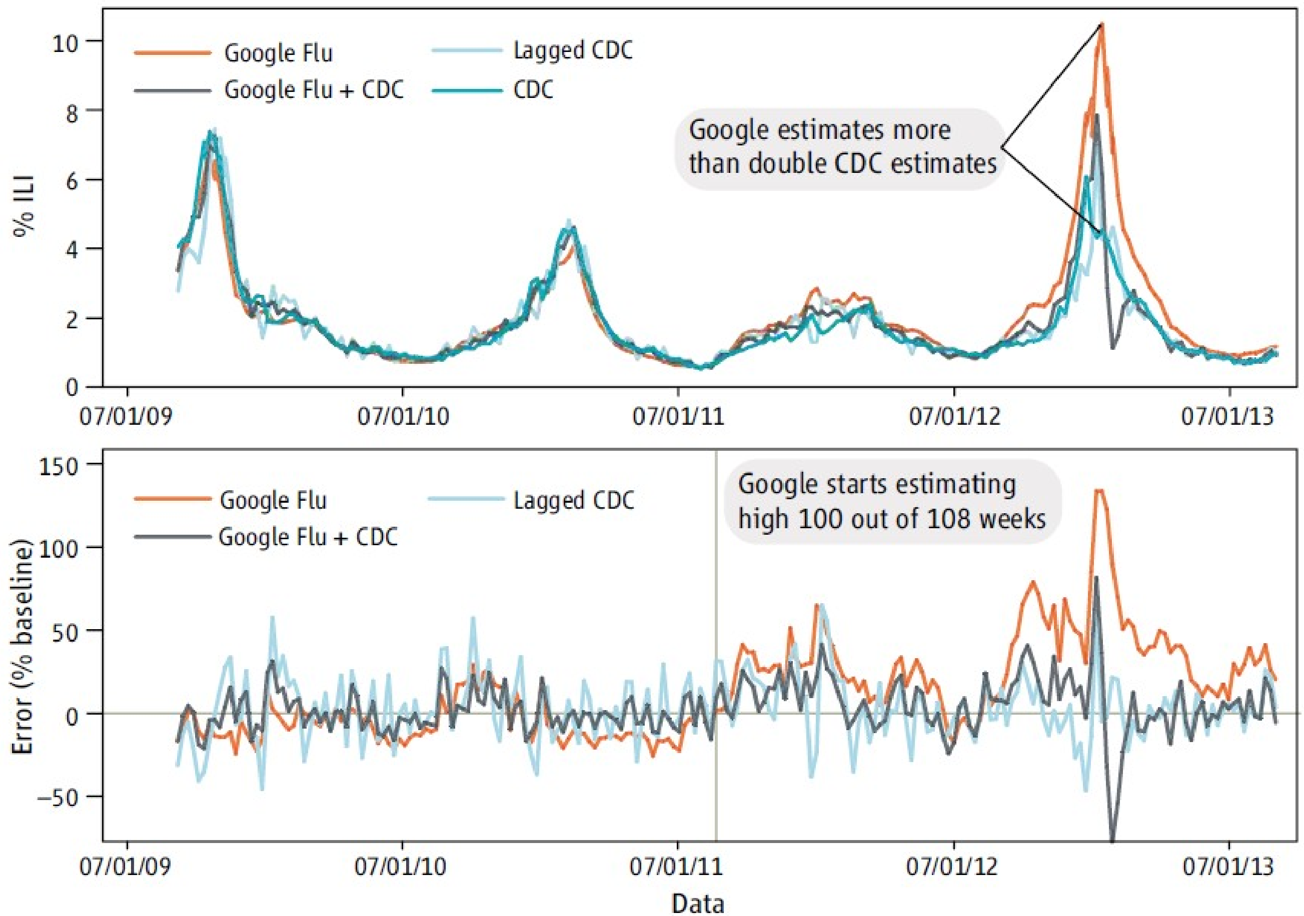
- **Incomplete data** (few variables)

Accuracy of Found Data

- Concept of error still applies
  - Estimate from found data can deviate from the parameter it's intended to estimate

- Concept of variance and bias still apply
  - Over repeated iterations, deviations can be random and/or systematic (caveats apply)

Real-time monitoring of flu (ILI: Influenza-like illness) using Google searches



Lazer, D., Kennedy, R., King, G., & Vespignani, A. (2014). The parable of Google Flu: traps in big data analysis. *Science*, *343*(6176), 1203-1205.

Why did it stop working?

- "Big Data hubris…occurs when the Big Data researcher believes that the volume of the data compensates for any of their deficiencies, thus obviating the need for traditional, scientific analytic approaches."
- Algorithm dynamics: Change in Google interface – started providing suggested search terms

Also:

- Overfitting – found searches that happened to be correlated with flu in one time period but not in another
- Increased media coverage of the flu in 2013

Concepts we've learnt…

"quantity of data does not mean that one can ignore foundational issues of measurement and construct validity and reliability." – Lazer et al. 2014

Algorithm Dynamics

All empirical research stands on a foundation of measurement. Is the instrumentation actually capturing the theoretical construct of interest? Is measurement stable and comparable across cases and over time? Are measurement errors systematic? At a minimum, it is quite likely that GFT was an unstable reflection of the prevalence of the flu because

https://gking.harvard.edu/files/gking/files/0314policyforumff.pdf

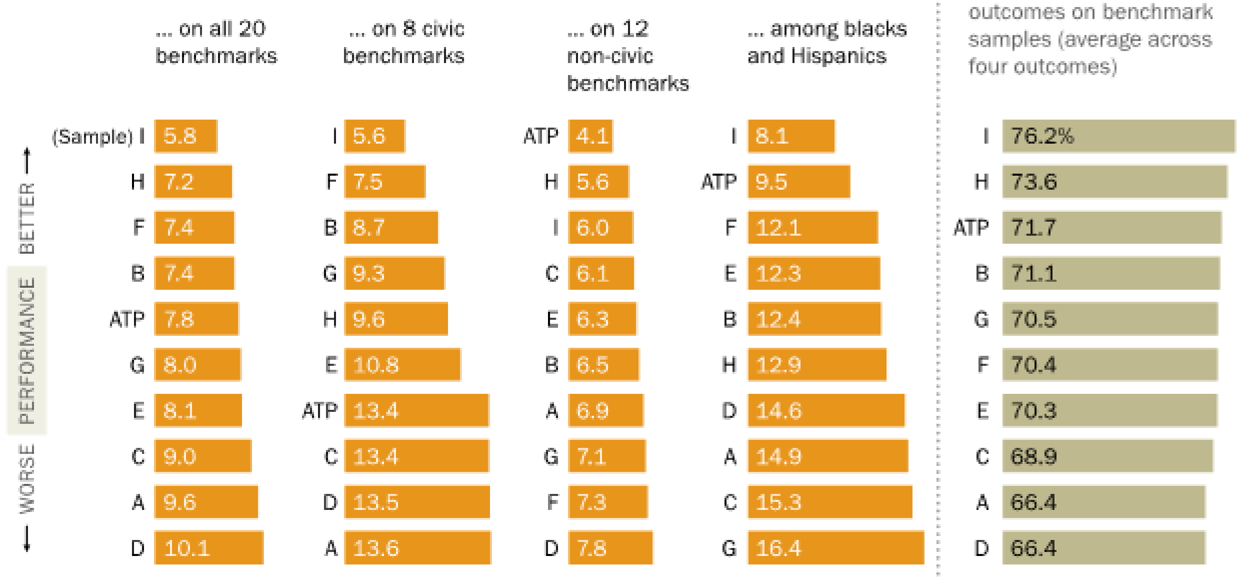
But a lot of good outcomes too…

Average estimated bias in benchmarking analysis ...

Values for each sample represent the average of the absolute differences between the population benchmarks and weighted sample estimates

Average % correctly classified in regressions

How well regression models from online samples predict outcomes on benchmark samples (average across four outcomes)



Note: Black and Hispanic averages exclude the driver's license benchmark which is not available for racial or ethnic subgroups. See Appendix D for details on individual benchmark items.

Source: Pew Research Center analysis of nine online nonprobability samples and the Center's American Trends Panel data. See Appendix A for details.
"Evaluating Online Nonprobability Surveys."

https://www.pewresearch.org/methods/2016/05/02/evaluating-online-nonprobability-surveys/

TSE for found Data

- TSE is based on data generating process of surveys, which may differ from data generating process for found data
- Traditional TSE does not include errors that are unique to found data
- Work in progress…

Select five proposed error frameworks …

1. Total Error Framework for a generic dataset

(Big Data Total Error (BDTE))

Record #	V_1	V_2	…	V_K
1				
2				
…				
N				

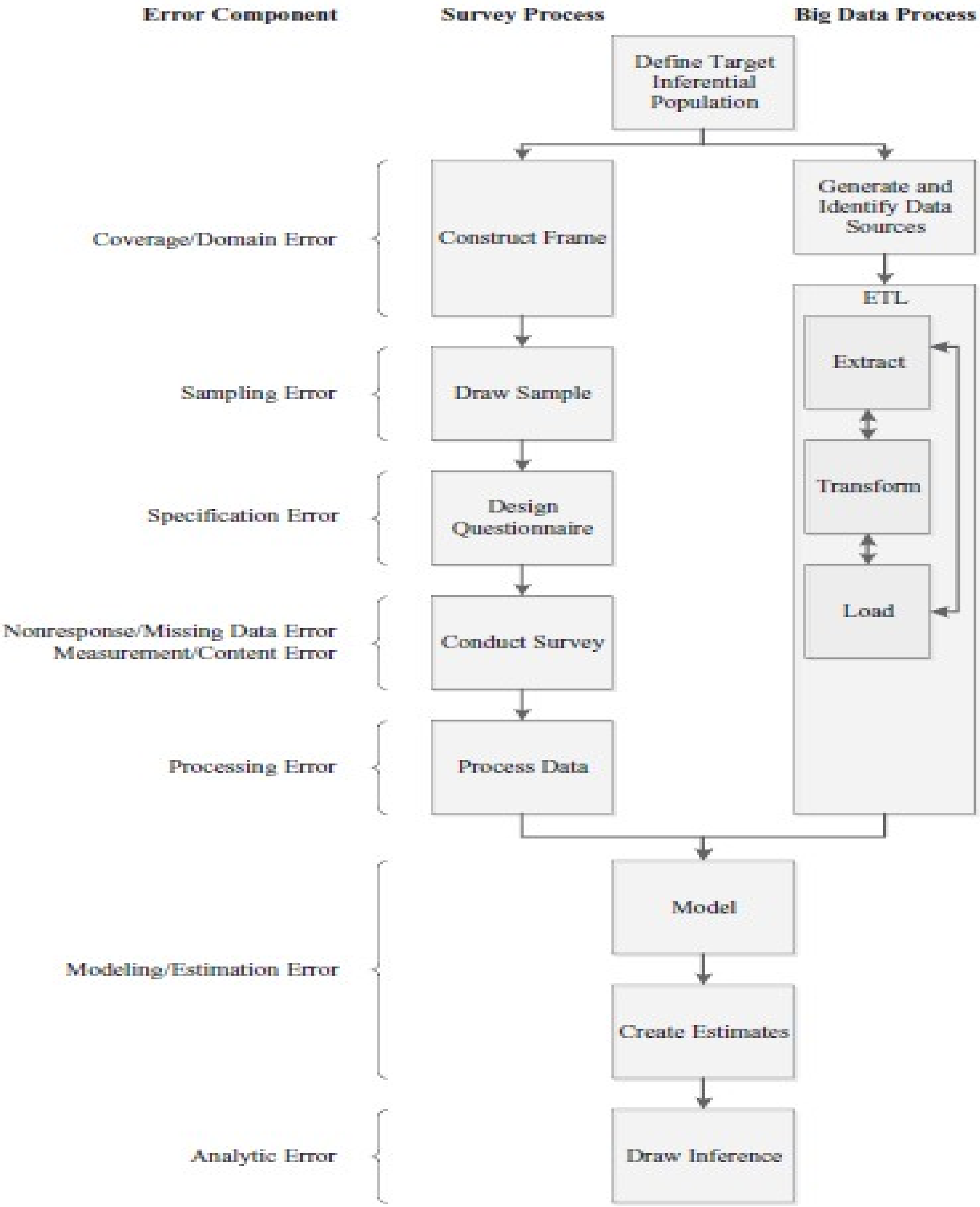
Rows -> Values corresponding to population units of some kind

Columns -> Characteristics of the population units

AAPOR (2015) – Big Data in Survey Research AAPOR Task Force Report
Biemer, P. P., & Amaya, A. (2020). Total error frameworks for found data. *Big Data meets survey science: a collection of innovative methods*, 131-161.

Total Error Framework…

- Total error = Row error + Column error + Cell error
- Row Error
  - Omission errors
  - Duplication errors
  - Erroneous inclusions
- Column Error
  - Misspecified variables (misaligned construct, outdated value, etc.)
- Cell Error
  - Content error: measurement error, linking error, data processing error, etc.
  - Missing data



2. Total Error Framework (TEF)

Amaya, A., Biemer, P. P., & Kinyon, D. (2020). Total error in a big data world: adapting the TSE framework to big data. *Journal of Survey Statistics and Methodology*, *8*(1), 89-119.

2015 Residential Energy Consumption Survey (RECS) and ‘big data’

- CAPI using ABS frame

- Can we just use ‘big data’ (e.g. Zillow) to estimate home square footage ?

RECS v/s Big data: Coverage error

- RECS: Presumably little coverage error
- Zillow: 110 Mn. Addresses. U.S. Census: 135 Mn. (coverage rate of 82%)
- 22% survey units unmatched to three ‘big data’ sources (processing error?).

Think: How would you estimate coverage bias?

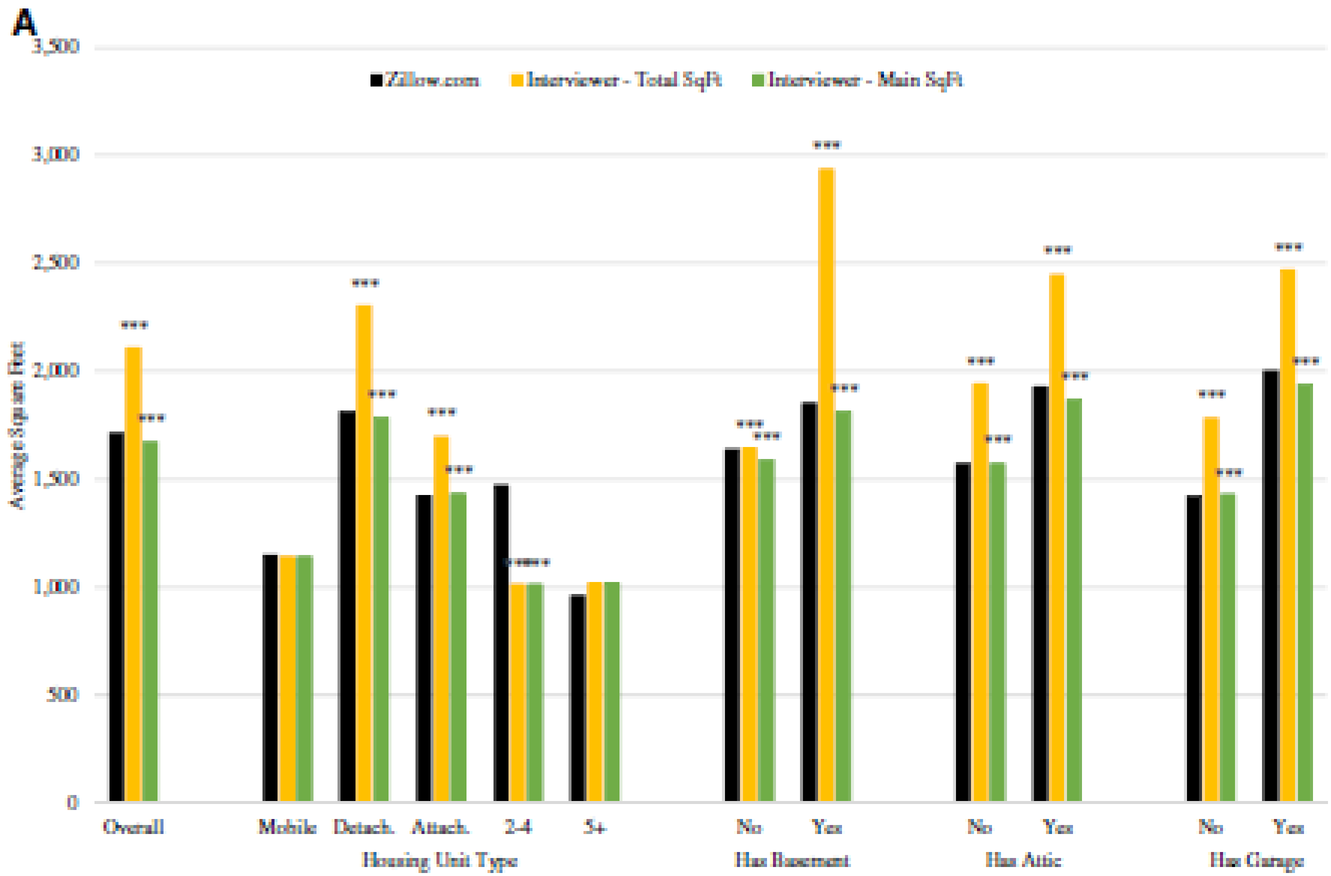
RECS and ‘big data’…

- Specification error (or is this measurement error?)

- Non-response error
  - 10% of Zillow records did not have housing unit size information

RECS and ‘big data’…

Measurement error



Risk profile…

TEF error component	Survey data			Big Data		
	Reason for error	Analysis	Assessed level of error	Reason for error	Analysis	Assessed levelof error
Coverage	Undercoverage	Qualitative evaluation of frame construction methods Qualitative inference given undercoverage conclusion	Minimal	Undercoverage	**Match rate** **Compared match rate by covariates of size and *t* test between matched and unmatched**	High
	Overcoverage	Qualitative evaluation of interviewer methods	Minimal	Overcoverage	Anecdotal evidence	Minimal
	Duplication	Qualitative evaluation of interviewer methods	Minimal	Duplication	Anecdotal evidence	Minimal

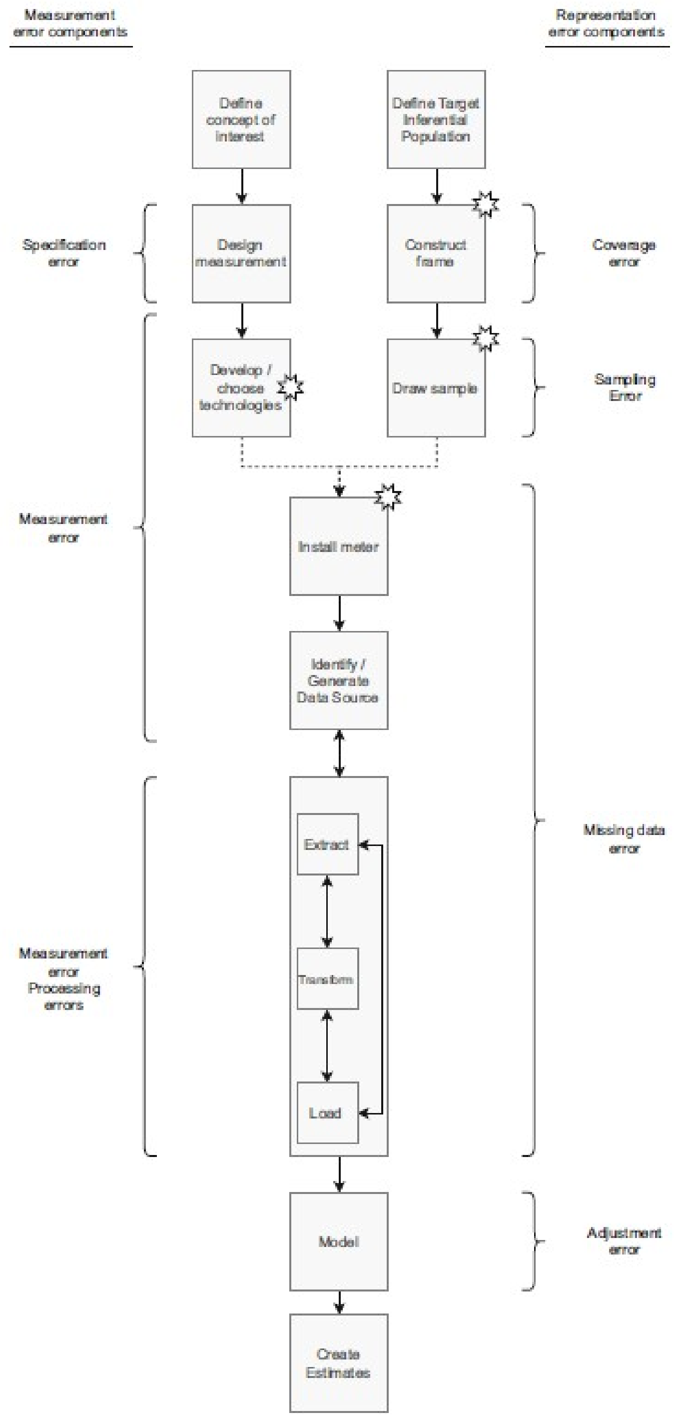
Also include other quality components

3. Total Error framework for digital traces collected with Meters (TEM) (Bosch and Revilla 2022)

- Metered data (web-tracking/weblog/digital trace data): Collected from a sample of participants who willingly install or configure, onto their devices, technologies that track digital traces left when people go online (e.g., URLs visited).
- Could be apps, browser plug-ins, proxies.

Metered data…

- Samples can be built using probability or non-probability sampling methods
- Focused on behaviors - no recall errors

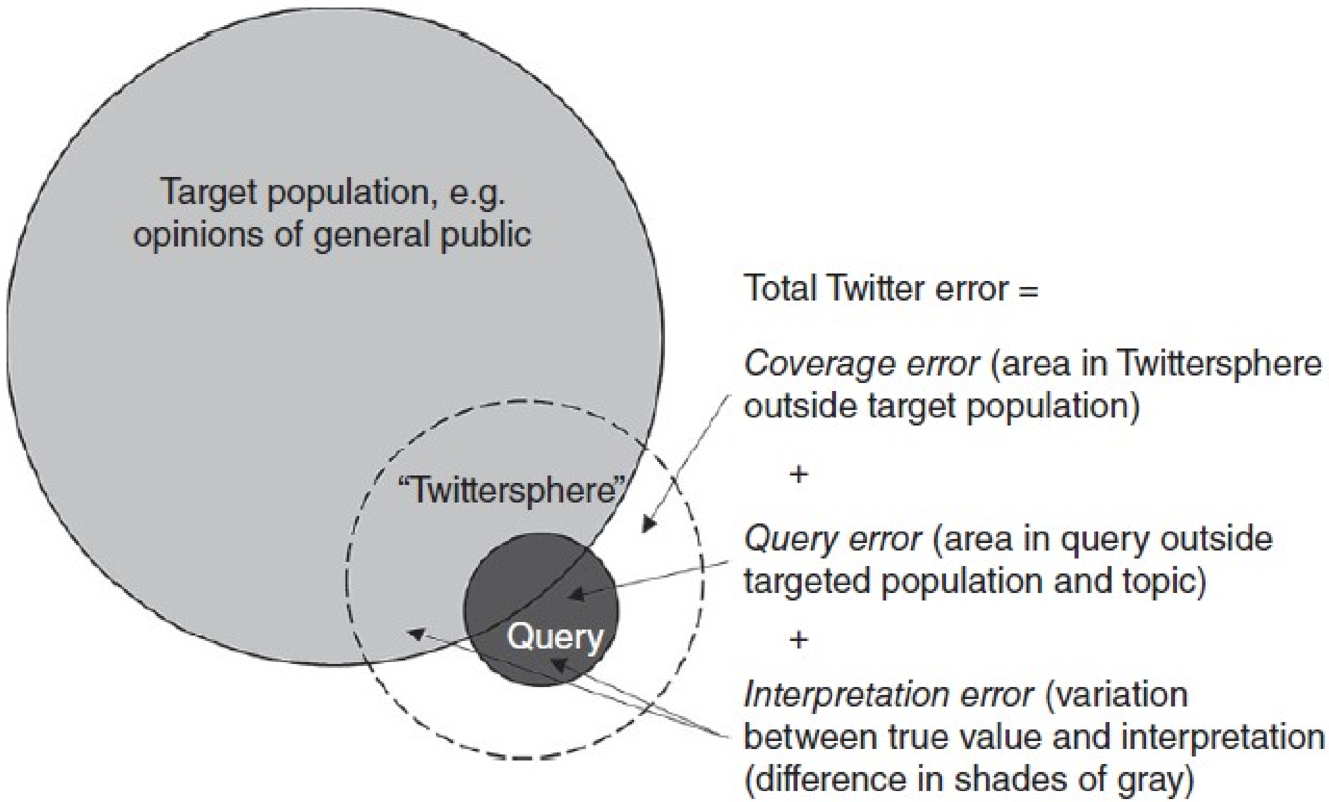


Measurement error components
Representation error components
Define concept of interest
Define Target Inferential Population
Specification error
Design measurement
Construct frame
Coverage error
Develop / choose technologies
Draw sample
Sampling Error
Measurement error
Install meter
Identify / Generate Data Source
Missing data error
Extract
Transform
Load
Measurement error Processing errors
Model
Adjustment error
Create Estimates

4. Total Twitter Error

Hsieh and Murphy, 2017

- Typical workflow of Twitter analysis:
  - Define geography, language, time frame
  - Define a set of topical keywords
  - Develop search query
  - Use automated techniques to analyze data
- Three error types…



Query example

- Basic query: *(marijuana OR pot) AND (legal OR legalize OR legalization)*
- Expanded query: *((marijuana OR pot OR 420 OR cannabis OR mmj OR weed OR hemp OR ganja OR THC) AND (legal OR legalize OR legalization)) OR mmot*

Coverage error

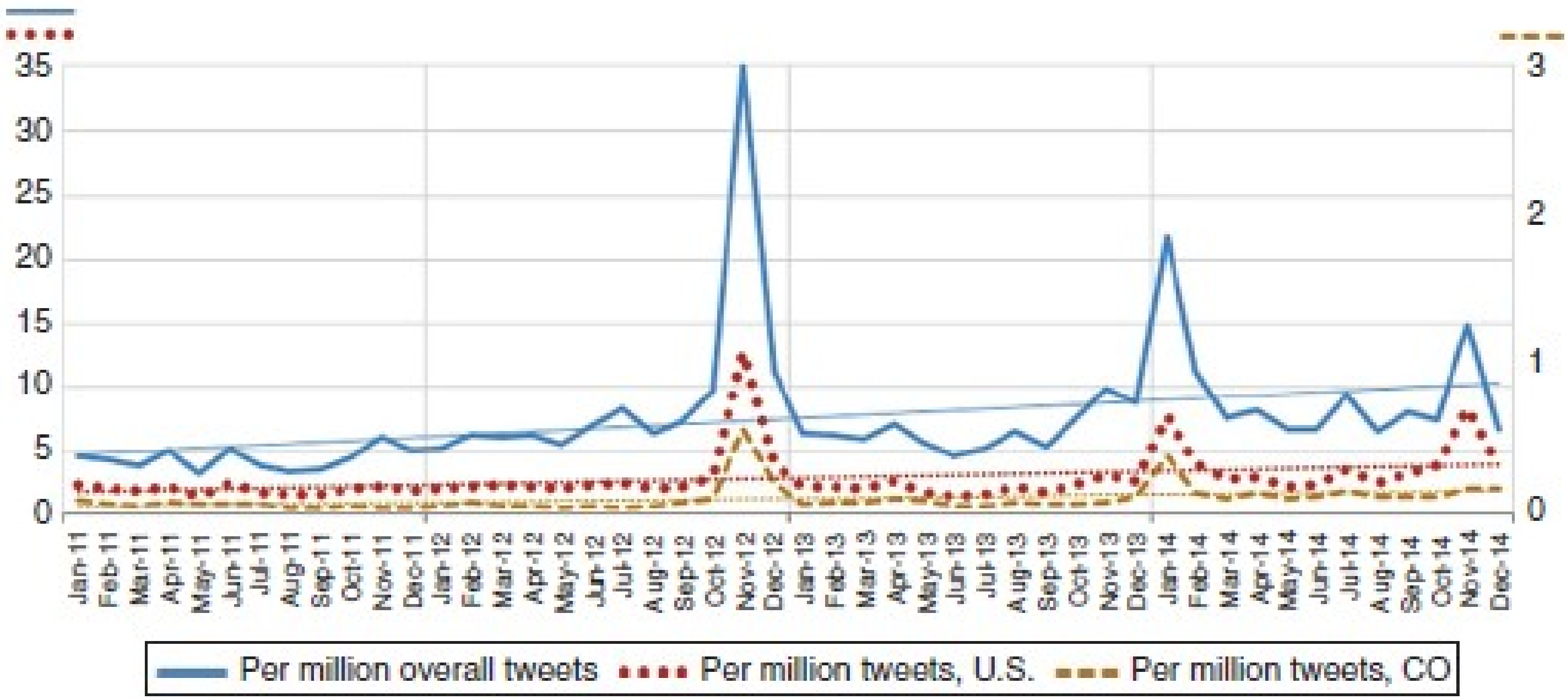


Figure 2.3 Marijuana legalization tweet volume by month, worldwide, U.S.A. and Colorado, basic query (2011–2014).

Why is the blue line not overlapping the dark brown line ?

Query error ?

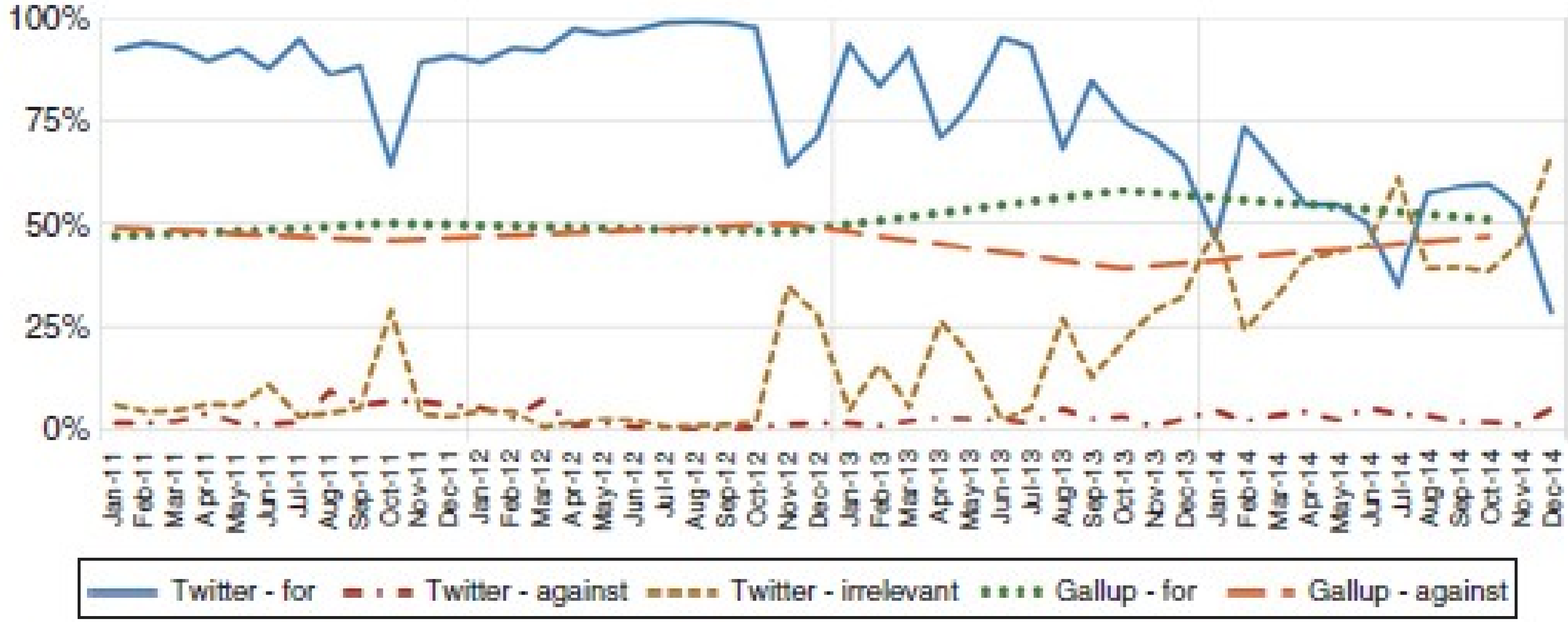


Figure 2.9 Marijuana legalization Twitter and Gallup sentiment by month, worldwide, basic query, 2011–2014.

- Dealing with links/images ?
- RTs – endorsement ?

Interpretation error

"*I Wanted to Predict Elections with Twitter and all I got was this Lousy Paper*"

A Balanced Survey on Election Prediction using Twitter Data

Daniel Gayo-Avello
dani@uniovi.es
@PFCdgayo
Department of Computer Science - University of Oviedo (Spain)

May 1, 2012

Abstract

Predicting X from Twitter is a popular fad within the Twitter research subculture. It seems both appealing and relatively easy. Among such kind of studies, electoral prediction is maybe the most attractive, and at this moment there is a growing body of literature on such a topic.

This is not only an interesting research problem but, above all, it is extremely difficult. However, most of the authors seem to be more interested in claiming positive results than in providing sound and reproducible methods.

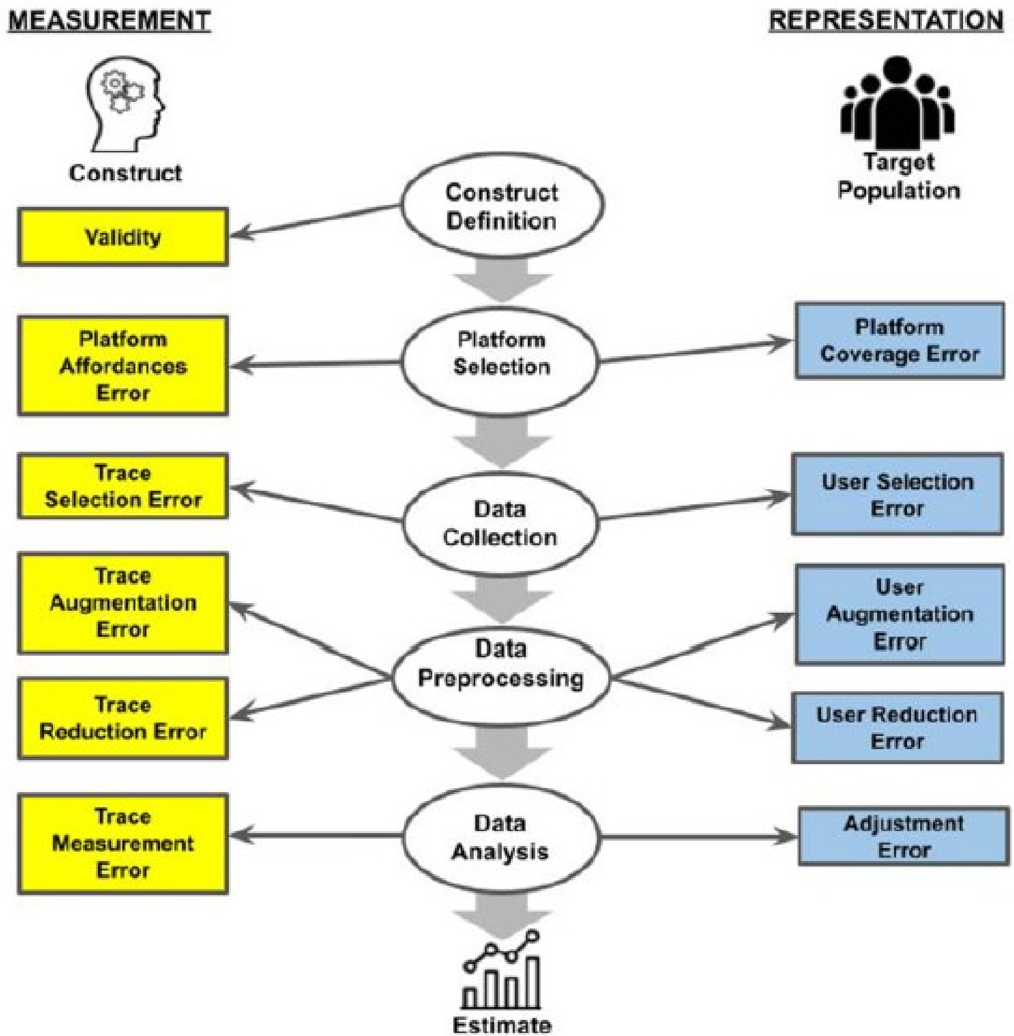
It is also especially worrisome that many recent papers seem to only acknowledge those studies supporting the idea of Twitter predicting elections, instead of conducting a balanced literature review showing both sides of the matter.

After reading many of such papers I have decided to write such a survey myself. Hence, in this paper, every study relevant to the matter of electoral prediction using social media is commented.

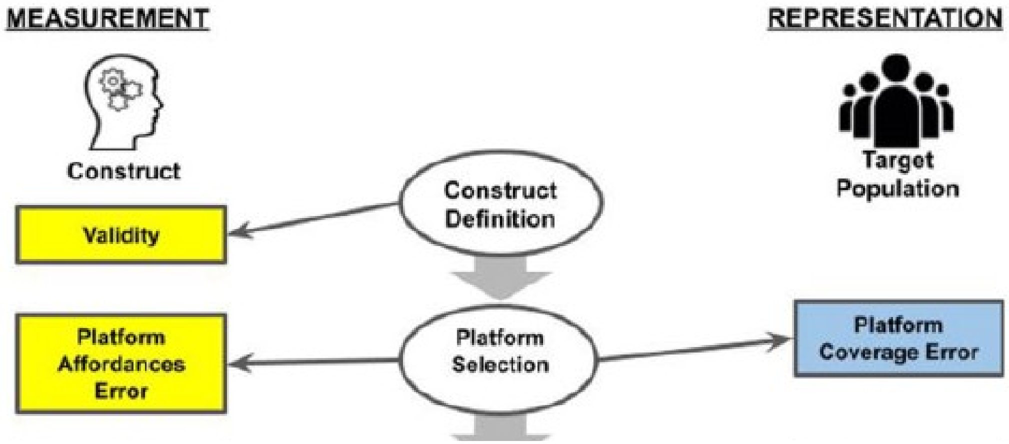
From this review it can be concluded that the predictive power of Twitter regarding elections has been greatly exaggerated, and that hard research problems still lie ahead.

5. Total Error Framework for Digital Traces of Human Behavior on Online Platforms (TED-On)

(Sen et al, 2021)



- ‘Traces’ and ‘users’ rather than ‘responses’ or ‘respondents’.
  - Users are ideally humans or institutions.
- “Trace”:
  - User-generated content
  - Interactions with content (e.g., liking, viewing, downloading)
  - Interactions between users (e.g., following)
- ‘Selection’ rather than ‘sample’

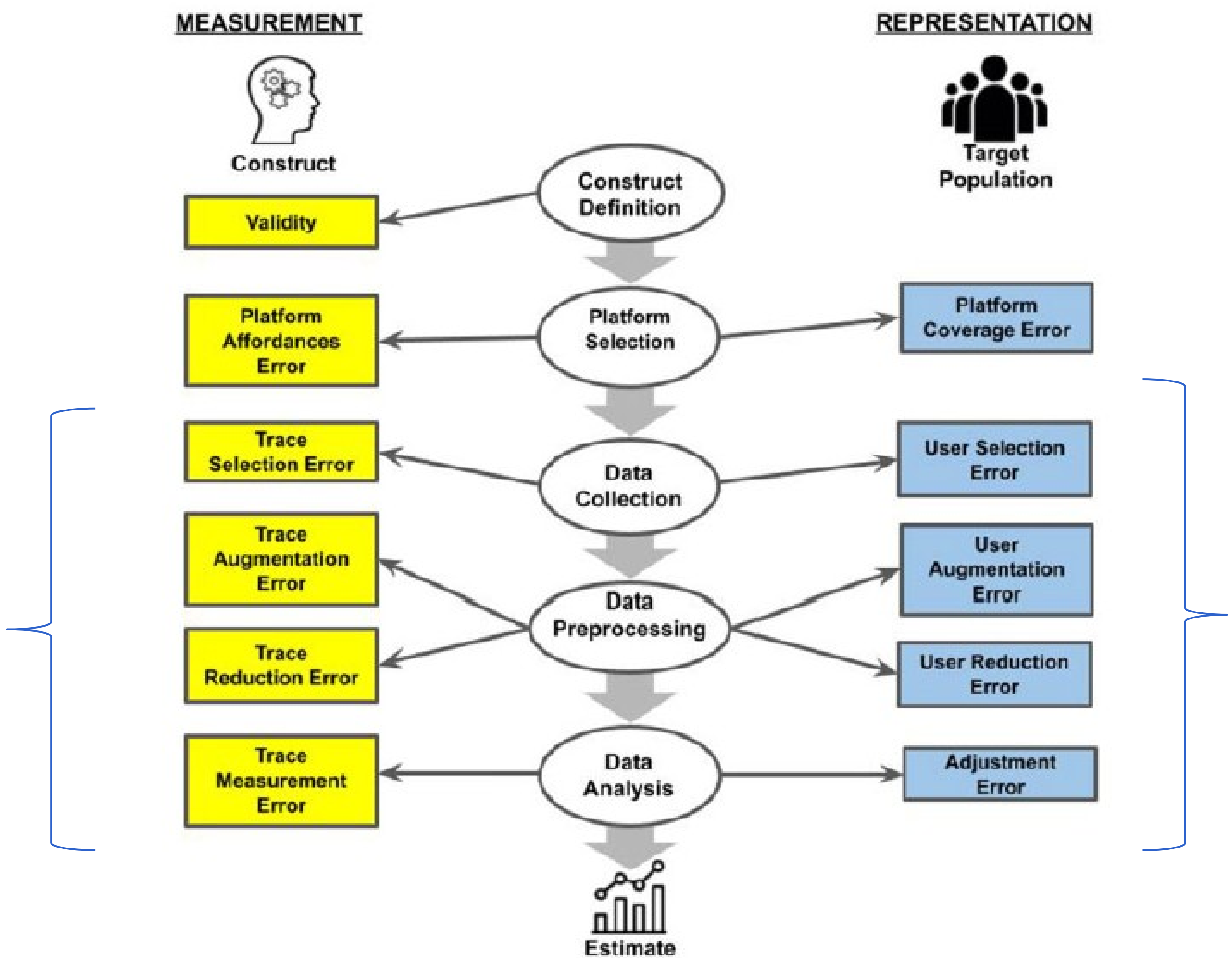


- Digital traces on the platform = theoretical construct of interest ?

- Platform population = target population ?

Platform affordances error

- Behaviors of users and traces influenced by:
  - Platform’s sociocultural norms (trending topics; ‘likes’; recommended connections; support base)
  - Design and Technical capability (e.g., word limit).Changes over time resulting in “behavioral drifts” impacting reliability over time
- Are platform-driven behaviors different from independent behaviors?



MEASUREMENT
REPRESENTATION
Construct
Target Population
Construct Definition
Validity
Platform Selection
Platform Affordances Error
Platform Coverage Error
Data Collection
Trace Selection Error
User Selection Error
Data Preprocessing
Trace Augmentation Error
Trace Reduction Error
User Augmentation Error
User Reduction Error
Data Analysis
Trace Measurement Error
Adjustment Error
Estimate

AI and Survey research

Some things AI can do for us…

- Automate the coding and analysis of open-ended responses

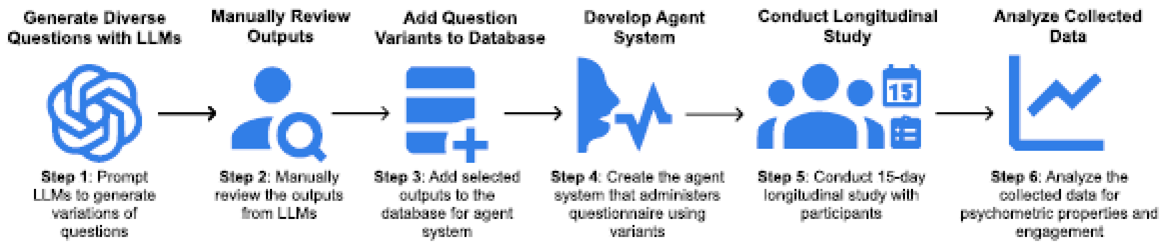
- Questionnaire Design

- Chatbots to help respondents

Increase respondent engagement

Yun et al 2024 *(https://arxiv.org/pdf/2311.12707)*

- Concern: Respondent fatigue in surveys
- Solution: Generate different questions targeting the same construct



- Longitudinal study: Compare responses from LLM-agent questions with survey responses. Ask respondents what they felt.
- Find good validity and reliability indices. Respondents find LLM less repetitive.

But need to be careful…

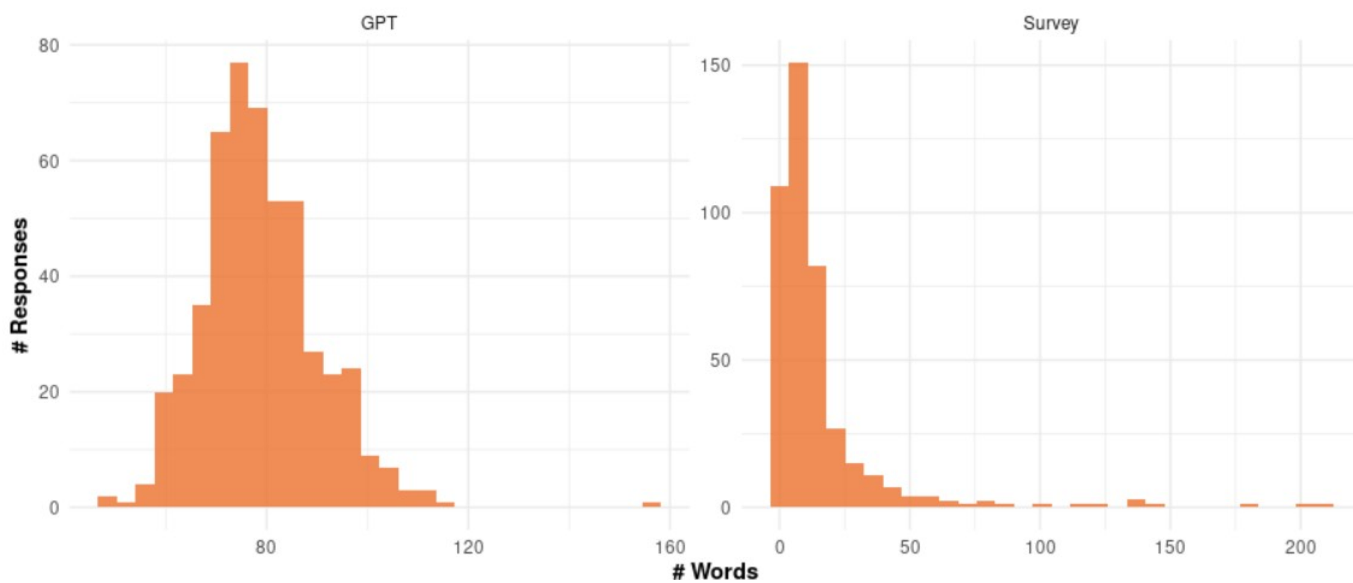
- LLMs are based on training data – can perpetuate these biases.

- Geng et al 2024:

(https://arxiv.org/abs/2405.19323)

Can large language models (LLMs) simulate social surveys? To answer this question, we conducted millions of simulations in which LLMs were asked to answer subjective questions. A comparison of different LLM responses with the European Social Survey (ESS) data suggests that the effect of prompts on bias and variability is fundamental, highlighting major cultural, age, and gender biases. We further

- Lerner et al (2024):
  - Use LLMs to generate synthetic responses to test different open-ended questions
  - Compare to human responses



https://aapor.confex.com/aapor/2024/meetingapp.cgi/Paper/3202

- Bisbee et al. (2023): ChatGPT-generated responses overly exaggerate the extremity and certainty of partisan and social division compared to actual opinions of those possessing the same attributes. (https://doi.org/10.31235/osf.io/5ecfa)

- Bisbee et al (2024): “Can a pretrained LLM produce synthetic opinions for respondent personas that accurately mirror what similar human respondents would say on a real survey ?”
  - Prompt ChatGPT to adopt different personas defined by political and demographic characteristics: then have it answer battery of questions about feelings toward social and political groups

Bisbee et al (2024)

"It is [YEAR]. You are a [AGE] year-old, [MARST], [RACETH] [GENDER] with [EDUCATION] making [INCOME] per year, living in the United States. You are [IDEO], [REGIS] [PID] who [INTEREST] pays attention to what's going on in government and politics."[13]

In each query to ChatGPT, the characteristics in brackets are substituted with values corresponding to a real respondent in the 2016 or 2020 wave of the ANES. These values include:

- [YEAR]: 2016 or 2020
- [AGE]: age in years of ANES respondent
- [RACETH]: non-Hispanic white, non-Hispanic black, or Hispanic
- [GENDER]: male or female
- [MARST]: divorced, married, separated, single, or widowed
- [EDUCATION]: a high school diploma, some college but no degree, a bachelor's degree or more
- [INCOME]: \$30k, \$50k, \$80k, \$100k, \$150k or more
- [IDEO]: an extremely liberal, a liberal, a slightly liberal, a moderate, a slightly conservative, a conservative, an extremely conservative
- [REGIS]: registered, unregistered
- [PID]: Democrat, Independent, Republican
- [INTEREST]: never, sometimes, frequently, regularly, always

"Provide responses from this person's perspective. Use only knowledge about politics that they would have. Format the output as a tsv table with the following format:

group,thermometer,explanation,confidence

The following questions ask about individuals' feelings toward different groups. Responses should be given on a scale from 0 (meaning cold feelings) to 100 (meaning warm feelings). Ratings between 50 degrees and 100 degrees mean that you feel favorable and warm toward the group. Ratings between 0 degrees and 50 degrees mean that you don't feel favorable toward the group and that you don't care too much for that group. You would rate the group at the 50 degree mark if you don't feel particularly warm or cold toward the group.

How do you feel toward the following groups?
The Democratic Party?
The Republican Party?
Democrats?
Republicans?
Black Americans?
White Americans?

….[cropped]

- Prompted 30 synthetic respondents for each of the 7,530 human respondents in the ANES survey, yielding a final dataset of 3,614,400 responses.

"Simply put, researchers cannot take for granted that responses from a pretrained LLM will match traditional survey data."

- 48% of coefficients estimated from the ChatGPT responses are statistically significantly different from their ANES-derived counterpart; among these cases, the sign of the effect flips 32% of the time.
- Distribution of results is sensitive to small differences in the prompt
- the distribution of responses to the same prompt changed between our initial run in April 2023 and a rerun in July 2023 due to changes in the underlying algorithm.

"Everyone wants to do the model work, not the data work":
Data Cascades in High-Stakes AI

Sambasivan et al, 2021

Here's where we definitely come in !

Key learning points from 720

- Different sources of error:
  - Causes
  - How to measure ?
  - How to reduce ?
- Good to start looking at different error sources independently but do not miss how they might interact. This is about Total Survey Error.
- TSE is one part of the total quality framework

Key learning points from 720…

- Costs are important
  - TSE tells us that there is often no free lunch

- Should be open to new methods/tech – but keep a critical eye

SURV/SURVMETH 721 preview

Next Semester

- Designed to walk you through the full process of conducting research AND writing an academic paper (including receiving and responding to reviews from peers).
  - Select some problem with a real survey
  - Conduct some analysis that sheds quantitative light on that source of error
    - How big is this error source?
    - What causes it?
    - What can be done about it?

Generating an idea

- How to generate an idea?
  - Different people use different approaches
- Can start with a research idea
  - e.g., investigate nonresponse bias in Web surveys
- Or start with a data set
  - public use data, data from work

Key Points

- Start thinking about this now; start generating ideas/looking for data sets.
  - Lot of freedom
- Do try to get your work published !

Coming up

- Course evaluations
- Dec 8: Final exam via Canvas
  - Available from 07:00 - 23:59 ET (login max by 21:59)
  - Structure similar to the mid-term
    - 2 hours
    - Open book, open notes
    - ALMOST fully focused on 2nd half (some concepts are just core)

"Survey research is a complicated sport and not well-suited for the faint of heart and especially not the untrained."

- Dever (2020)

Journal of Survey Statistics and Methodology (2020) 8, 433–441

I hope 720 helps you play well !

Thank you!